

Tools & Trade:

1. X64dbg - For patching
2. C2 (command and control server) of your choice.
3. Shellcode generator of your choice
4. Software to be patched of your choice (Each students must select different software, Write the software name & link in list and your Sr Number and

Hint: target a small software, it will be easy to handle it.

5. Microsoft Windows(10/11)
-

Reverse Engineering, Patching & C2

Part 1:

Note: After reading all instructions, it should be clear to you that this is static patching so a patched version of executable will/should be produced at the end of this project.

1. Choose a Closed Source Application (this does not include your written programs), and write it in the given list. Do not put your shellcode in source code directly, you are to strictly work using reverse engineering methodology & patching in the main executable of the software. You must not call any external file (means your malicious code will reside inside the executable).
2. Reverse engineer it to include your malicious code for proof of concept in the executable of the application of your choice. (For Part 1 , you must choose a simple shell-code to do something like pop-up notepad,calculator or anything.).

Part 1 shellcode must be different from Part 2.

Note: you may use meterpreter(msfvenom),

<https://www.exploit-db.com/shellcodes> or any shellcode/shellcode generator of your choice. But the action of your shellcode must be visible. Crashing of program is a visible but not a malicious action so its not eligible for credit)

Part 2:

1. Select the C2 server of your choice (Few popular ones are : Covenant c2, Sliver, Havoc, Mythic and many more. It is advised to do research and select the C2 you think best suits you).
 2. Generate the shellcode for reverse (TCP) connection to your C2 server.
 3. Patch your selected software with the malicious shellcode you generated.
 4. Provide a proof of concept by executing the patched software and a maintained connection with the C2 server.
 5. With Anti-Virus (MicroSoft Defender)disabled, you will be eligible for 95% marks
 6. For full credit, your malicious activity should not be detected even with MS Defender enabled.
-